

ARTIFICIAL INTELLIGENCE

Chapter 1: Introduction

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Outline

- What is Artificial Intelligence (AI)?
- The foundations of AI
- The history of AI
- The state of the art

Artificial Intelligence



Artificial Intelligence

- Homo sapiens – man the wise
- How we think i.e.
 - Perceive, understand, predict & manipulate
- AI is not just to understand but build intelligent entities
- Field I would most like to be in

What is AI?

Views of AI fall into four categories:

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

The textbook advocates "acting rationally"

Thinking Humanly

“The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.” (Haugeland, 1985)

“[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)

Acting Humanly

“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990)

“The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)

Thinking Rationally

“The study of mental faculties through the use of computational models.”
(Charniak and McDermott, 1985)

“The study of the computations that make it possible to perceive, reason, and act.”
(Winston, 1992)

Acting Rationally

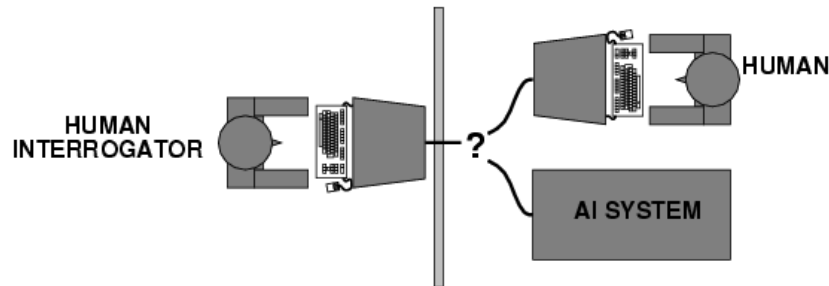
“Computational Intelligence is the study of the design of intelligent agents.” (Poole *et al.*, 1998)

“AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Acting humanly: Turing Test

- The **Turing Test**, proposed by Alan Turing (1950)
- "Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game



- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Suggested major components of AI: knowledge, reasoning, language understanding, learning
- Total Turing test with video signal: computer vision, robotics

Thinking humanly: cognitive modeling

- Requires scientific theories of internal activities of the brain
- How to validate?
 - introspection—trying to catch our own thoughts as they go by
 - psychological experiments—observing a person in action
 - brain imaging—observing the brain in action
- The interdisciplinary field of **cognitive science** brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

Thinking rationally: "laws of thought"

- Aristotle was one of the first to attempt to codify “right thinking,”
- Provided patterns for argument structures that always yielded correct conclusions when given correct premises
- for example,
 - “Socrates is a man; all men are mortal; therefore, Socrates is mortal.”
- Problems:
 - not easy to convert informal knowledge into formal (logical notation), when the knowledge is less than 100% certain.
 - Second, there is a big difference between solving a problem “in principle” and solving it in practice(exhaust the computational resources)

Acting rationally: rational agent

- **Rational** behavior: doing the right thing
- A **rational agent** is one that acts so as to achieve the **best outcome** or, when there is uncertainty, the **best expected outcome**.
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action
- Advantages
 - more general
 - amenable to scientific development

Rational agents

- An **agent** is an entity that perceives and acts
 - operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals.
- Abstractly, an agent is a function from percept histories to actions:
$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$
- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance
- Caveat: computational limitations make perfect rationality unachievable
 - design best **program** for given machine resources

The foundations of AI

❖ Philosophy

- Can formal rules be used to draw valid conclusions?
- How does the mind arise from a physical brain?
- Where does knowledge come from?
- How does knowledge lead to action?
- **Syllogisms**
 - an instance of a form of reasoning in which a conclusion is drawn from two given or assumed propositions (premises); deductive reasoning
- **Rationalism**
 - power of reasoning in understanding the world
- **Dualism**
 - a part of the human mind (or soul or spirit) that is outside of nature, exempt of physical laws
- **Materialism**
 - brain's operation according to the laws of physics constitutes the mind
- **Empiricism**
 - principle of **induction**: that general rules are acquired by exposure to repeated associations between their elements.
- **Aristotle Algorithm**
 - actions are justified by a logical connection between goals and knowledge of the action's outcome e.g. General Problem Solver now Regression Planning System.

The foundations of AI

❖ Mathematics

- What are the formal rules to draw valid conclusions?
- What can be computed?
- How do we reason with uncertain information?

• Logic

- Propositional Logic, First Order Logic
- Algorithm
 - Logical deduction
- Incompleteness theorem
 - some functions on the integers cannot be represented by an algorithm—that is, they cannot be computed.

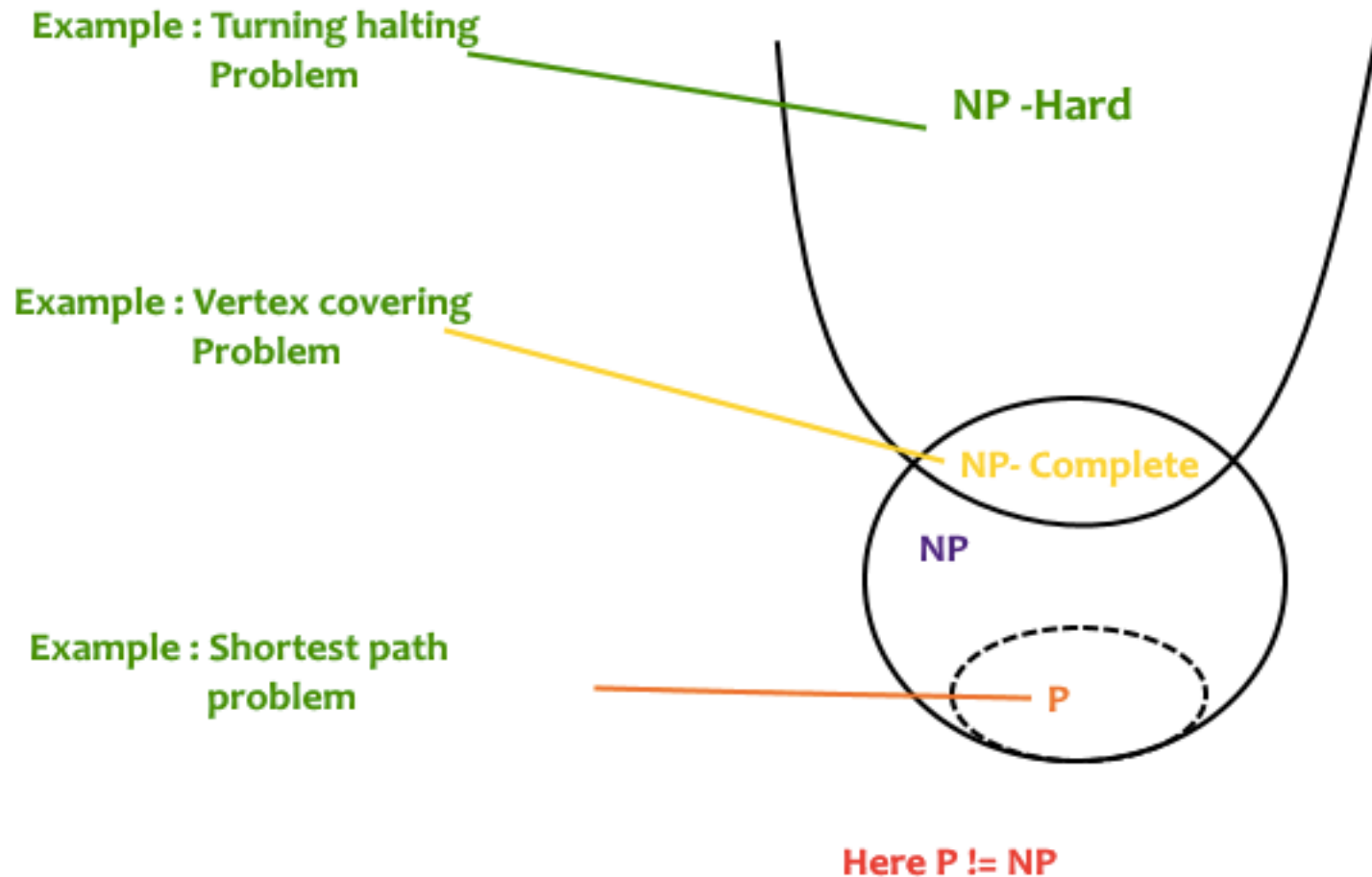
• Computation

- Tractability
 - a problem is called intractable if the time required to solve instances of the problem grows exponentially with the size of the instances
- NP-completeness
 - Intractable problem are NP-Complete

• Probability

- Bayesian Rule
 - updating probabilities in the light of new evidence

NP Complete



The foundations of AI

❖ Economics

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in the future?

• Utility

- how people make choices that lead to preferred outcomes or utility

• Decision theory

- probability theory + utility theory
- framework for decisions (economic or otherwise) made under uncertainty

• Game theory

- the actions of one player can significantly affect the utility of another (either positively or negatively).

• Operations research

- Markov decision processes

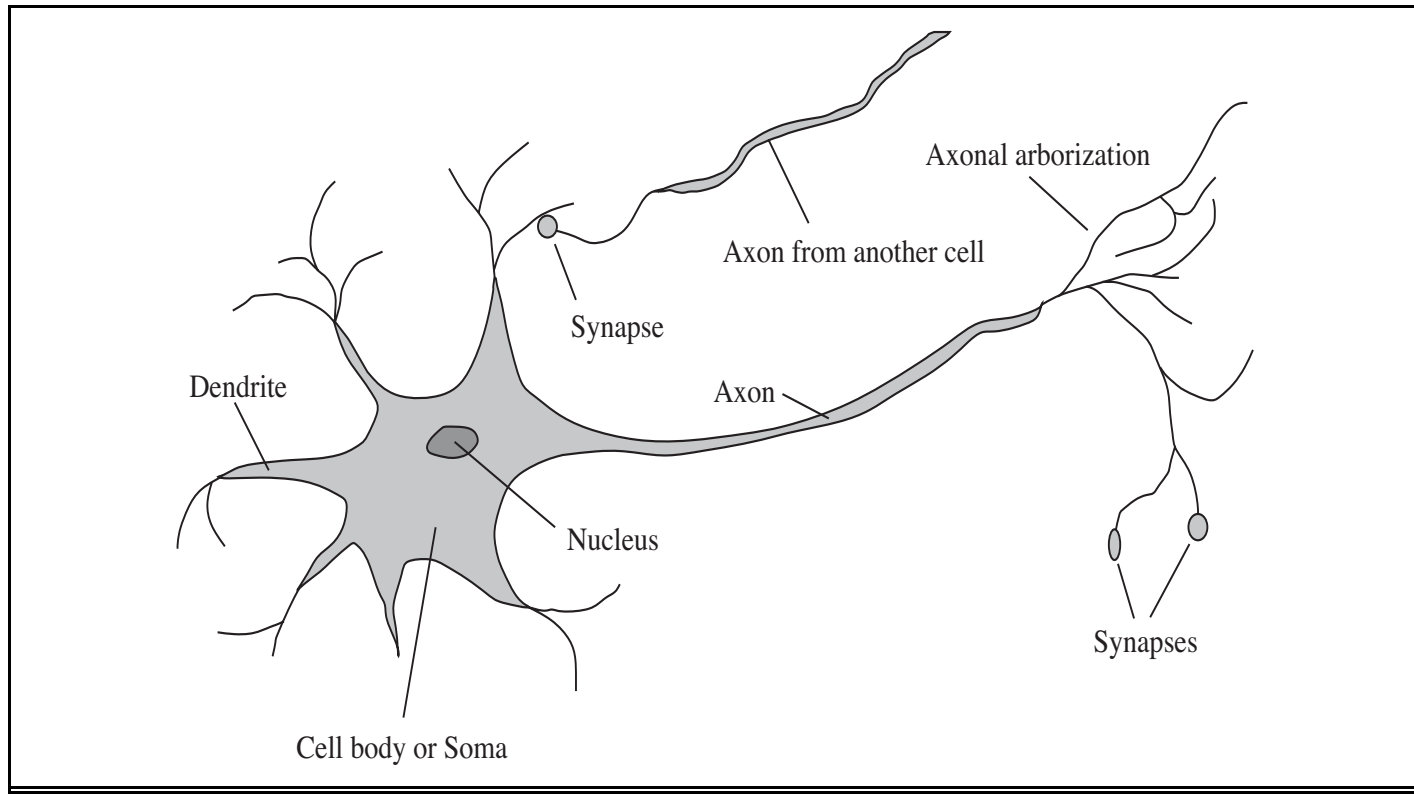
• Satisficing

- making decisions that are “good enough”

The foundations of AI

❖ Neuroscience

- How do brains process information?



The foundations of AI

❖ **Psychology**

- How do humans and animals think and act?

• **Behaviorism**

- Study only percepts and actions, not mental processes

• **Cognitive psychology**

- views the brain as an information-processing device

• **Cognitive science**

- how computer models could be used to address the psychology of memory, language, and logical thinking

The foundations of AI

❖ **Computer engineering**

- How can we build an efficient computer?
- For artificial intelligence to succeed, we need two things: intelligence and an artifact.
- **Programmable machine**
- **Computer Languages**
- **Time sharing machines**
- **Single Core to Multi Core**

The foundations of AI

❖ **Control theory and cybernetics**

- How can artifacts operate under their own control?

- **Control theory**

- self-regulating feedback control systems include the steam engine governor, and the thermostat
- regulatory mechanism trying to minimize “error”—the difference between current state and goal state.

- **Cybernetics**

- Study of the possibility of artificially intelligent machines

- **Homeostatic**

- devices containing appropriate feedback loops to achieve stable adaptive behavior.

- **Objective function**

- The goal is to design of systems that maximize an **objective function** over time. This roughly matches our view of AI: designing systems that behave optimally.

The foundations of AI

❖ Linguistics

- How does language relate to thought?
- **Computational linguistics** or **natural language processing**.
- Understanding language requires an understanding of the subject matter and context, not just an understanding of the structure of sentences.

A bridged history of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1952—69 Look, Ma, no hands!
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1965 Robinson's complete algorithm for logical reasoning
- 1966—73 AI discovers computational complexity
Neural network research almost disappears
- 1969—79 Early development of knowledge-based systems
- 1980-- AI becomes an industry
- 1986-- Neural networks return to popularity
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents
- 2001-- The availability of very large data sets

State of the art

- **Robotic vehicles**

STANLEY (2005), DARPA, CMU's BOSS (2006) in urban settings, Google Self-Drive, Tesla Autopilot, Uber

- **Speech recognition**

Book a flight through automated speech recognition and dialog management system.

- **Autonomous planning and scheduling**

NASA's Remote Agent program, MAPGEN (NASA's Mars Exploration Rovers), MEXAR2 (European Space Agency's Mars Express mission)

- **Game playing**

IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov

State of the art

- **Spam fighting**

Learning algorithms classify over a billion messages as spam, for many users, could comprise 80% or 90% of all messages, it is difficult for static programs to tackle the changing tactics of spammers.

- **Logistics planning**

In Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART to do automated logistics planning and scheduling for transportation. This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.

State of the art

- **Robotics**

IRobot sold over two million Roomba robotic vacuum cleaners for home use.

- **Machine Translation**

A computer program automatically translates from Arabic to English, etc. The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling two trillion words.

- **ChatGPT: Advancements in Conversational AI**

ChatGPT is a chatbot and virtual assistant developed by OpenAI and launched on November 30, 2022. Based on large language models (LLMs). It has over 100 million users.